MIL-S-43828A
7 January 1983
SUPERSEDING
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22 January 1973

MILITARY SPECIFICATION

STRAP, WEBBING, CARGO TIE DOWN, LIGHTWEIGHT PACK FRAME, M-1972

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This document covers the requirements for a strap consisting of nylon webbing, a metal fastener and a metal hook.

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein:

SPECIFICATIONS

FEDERAL

V-T-285	-	Thread, Polyester
UU-P-268	-	Paper, Kraft, Wrapping
DDD-L-20	-	Label; for Clothing, Equipage, and Tentage, (General Use)
PPP-B-636	-	Boxes, Shipping, Fiberboard

MILITARY

MIL-E-480	-	Enamel, Baking, Phenol - Or Urea-Formaldehyde
MIL-C-13924	-	Coating, Oxide, Black, For Ferrous Metals
MIL-W-43668	-	Webbing, Textile, Bulked Nylon

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Laboratories, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8465

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STANDARDS

FEDERAL

FED-STD-751 - Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment

DRAWINGS

US ARMY NATICK RESEARCH AND DEVELOPMENT LABORATRIES

2-2-313 - Strap, Webbing, Cargo Tie Down, Lightweight Pack Frame, M-1972

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B 633 - Electrode Posited Coatings of Zinc on Iron and Steel

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 Guide sample. Guide samples of the end item, when furnished, are solely for guidance and information to the contractor. Variations from this document may appear in the sample, in which case this document shall govern.

3.2 First article. When specified, a sample shall be subjected to first article inspection (see 4.3, 6.2 and 6.4).

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3.3 Materials. Materials specified shall conform to the applicable documents and requirements specified herein (see 6.5).

3.3.1 Webbing, nylon. The one inch nylon webbing shall conform to type III, Olive Drab 7 of MIL-W-43668.

3.3.2 Thread, polyester. The thread shall be type I, class 1, sub-class A or B, size F of V-T-285. When sub-class B thread is used the non-wicking requirements shall not apply.

3.3.2.1 Color. The thread shall be dyed Olive Drab S-1 (C.A. 66022) and shall show fastness to weathering equal to or better than the standard sample.

3.3.3 Fastener, steel. The steel fastener shall conform to Arizona Gear and Manufacturing Co., Part No. K415, Kev-Lok; Acorn Metal Products Co., Part Nos. C-103 and A-101; G. E. Prentice Manufacturing Co., Part No. 3837, or equal, (see 6.6). Prior to the use of the "or equal" item, the contractor shall submit the item to the contracting officer with supporting data for approval.

3.3.3.1 Finish. The rivets shall be given a black oxide coating conforming to class 2, 3 or 4, as applicable, of MIL-C-13924. The two lever components of the fastener shall be given a phosphate treated zinc plate conforming to type IV (with a minimum thickness of 13 micrometers) of ASTM B 633 followed by baked enameling. The bake enamel shall conform to type I or II of MIL-E-480, color black. The coating on all components shall be performed prior to assembly and shall level out to a smooth and uniform film without orange peel, wrinkles, drops, streaks or areas of no film (see 4.4.1.1).

3.3.4 Hook, steel. The steel hook shall conform to Arizona Gear and Manufacturing Co. Part No. K465; Acorn Metal Products Co. Part No. B-102; or G. E. Prentice Manufacturing Co. Part No. 3837 or equal (see 6.6). Prior to the use of the "or equal" item, the contractor shall submit the item to the contracting officer with supporting data for approval. The hooks used with the fasteners specified in 3.3.3 shall be made by the fastener manufacturer.

3.3.4.1 Finish. The finish for the hook shall be as specified in 3.3.3.1 for the fastener lever components (see 4.4.1.1).

3.4 Construction. The construction of the strap shall conform in all respects to Drawing 2-2-313 and as specified herein.

3.4.1 Stitching, machine. All stitching except bartacking shall conform to FED-STD-751, type 301 with 6 to 8 stitches per inch.

3.4.1.1. Type 301 stitching. Ends of all stitching shall be backstitched or overstitched not less than 1 inch. Thread tension shall be maintained so that there will be no loose stitching resulting in loose bobbin or top thread, or excessively tight stitching resulting in puckering of the material sewn. The lock shall be embedded in the materials sewn.

3.4.1.1.1 Repairs of type 301 stitching. Repairs of type 301 stitching shall be as follows:

a. When thread breaks or bobbin run-outs occur during stitching, the stitching shall be repaired by restarting the stitching a minimum of one inch (1/2 inch for W-W stitching) back of the end of the stitching. 1/

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b. Thread breaks, or two or more consecutive skipped or run-off stitches noted during inspection of the item (in-process or end item) shall be repaired by overstitching. The stitching shall start a minimum of one inch in back of the defective area, (1/2 inch for W-W stitching) continue over the defective area and continue a minimum of one inch (1/2 inch for W-W stitching) beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching, without damaging the materials, and restitching in the required manner. 1/

1/ When making the above repairs the ends of the stitching are not required to be backstitched.

3.4.1.2 Automatic stitching. Automatic stitching machines may be used to perform any of the stitch patterns provided the requirements for the stitch pattern, stitches per inch, and size and type of thread are met; and at least three tying, overlapping or backstitches are used to secure the ends of the stitching.

3.4.1.3 Bartacking. Bartacks shall be 3/4 ± 1/8 inch long, 1/8 ± 1/32 inch wide and shall contain 42 stitches. Bartacks shall be free from thread breaks and loose stitching.

3.4.1.4 Thread ends. All thread ends shall be trimmed to 1/4 inch maximum length.

3.4.2 Fusing of nylon webbing ends. All ends of nylon webbing shall be fused. The apparatus used to fuse the webbing ends shall be capable of providing sufficient heat to provide a smooth edge with the cut ends of the webbing yarns all fused together.

3.5 Marking. The identification marking shall conform to type IV, class 5 of DDD-L-20 and shall be set in two lines and located as shown in Drawing 2-2-313. The letters "US" shall be applied in the location and in the size characters indicated on the drawing and shall conform to type IV, class 9 of DDD-L-20. Fastness of the class 9 marking shall be as specified for class 5 markings.

3.6 Replacement of defective components. During the spreading, cutting and manufacturing process, components having material defects or damages that are classified as defects in 4.4.2.1 and 4.4.2.2 shall be removed from production and replaced with non-defective and properly matched components.

3.7 Workmanship. The finished strap shall conform to the quality of product established by this document. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or

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purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.2.1 and 4.4.2.2. The presence of any defect shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents, unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.4.1.1 Certification. The supplier shall furnish a certificate of compliance for the finish specified in 3.3.3.1 for the steel fastener and in 3.3.4.1 for the steel hook.

4.4.2 End item examination. The end item shall be examined in accordance with 4.4.2.1 and 4.4.2.2. The lot size shall be expressed in units of one strap. The sample unit shall be one strap.

4.4.2.1 Visual examination. The end item shall be examined for the defects listed below. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 1.5 for major defects and 4.0 for total (major and minor combined) defects.

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Examine	Defect	Classification Major	Minor
Webbing	Not firmly and tightly woven; edges frayed or scalloped	X	
	Multiple floats		X
	Any cut, hole, tear, or smash	X	
	Abrasion mark, slub, broken end or pick		X
	Ends not fused as specified (where required)		X
Metal hardware (general)	Broken or malformed failing to serve intended purpose, corroded area, burr, or sharp edge	X	
	Finish omitted, partially omitted, or not as specified: on steel components		X
	Not assembled as specified (unless otherwise classified herein)	X	
	Not specified type, size or style	X	
Stitching	Missing, not specified pattern	X	
Stitch type	Wrong seam or stitch type	X	
Stitch tension	Loose, resulting in a loose bobbin or top thread		X
Thread breaks, skipped stitches, or runoffs	Not overstitched as specified		X
NOTE:	Thread breaks or two or more consecutive skipped or runoff stitches not overstitched		X
Ends of stitching	Not secured as specified		X
Thread ends	Not trimmed to 1/4 inch maximum length		X
Components and assembly	Any component part omitted or not as specified or any required operation omitted (unless otherwise classified herein)	X	
	Needle chews	X	
	Any mend, darn, patch or splice	X	

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Examine	Defect	Classification Major	Minor
Marking	Omitted, illegible, incorrect, misplaced, or not as specified		X
Cleanness	Grease, ink, or oil stains clearly noticeable		X

4.4.2.2 Dimensional examination. Examination shall be made for compliance with all dimensions shown on the drawing which can be examined on the end item, excluding reference dimensions. Any dimension exceeding the applicable tolerance shall constitute a defect. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

4.4.3 Packaging inspection. An examination shall be made to determine that preservation, packing, and marking requirements comply with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully packaged with the exception that it need not be closed. Examination of closure defects listed below shall be made on shipping containers fully packaged. The lot size shall be the number of containers in the inspection lot. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing. Any component not as specified. Any component damaged.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, improper taping, loose strapping, or inadequate stapling. Bulged or distorted container.
Content	Number of straps per shipping container is more or less than required. 1/

1/ For this defect, the number of straps in one shipping container in the sample shall be examined.

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4.4.4 Palletization examination. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully packaged. The lot size shall be the number of palletized unit loads in the end item inspection lot. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement.
Palletization	Not as specified. Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. Each strap shall be compactly folded or rolled and securely tied with cotton tape or twine.

5.1.2 Commercial. Straps shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A packing. Three hundred straps, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class weather resistant, variety DW, grade V15c of PPP-B-636. Each container shall have the contents completely covered on the top and bottom with a sheet of 30-pound minimum basis weight kraft paper conforming to type I grade B of UU-P-268. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636, except the inspection shall be in accordance with 4.4.3.

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5.2.2 Level B packing. Three hundred straps, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class domestic, variety DW, grade 275 of PPP-B-636. Each container shall have the contents completely covered on the top and bottom with a sheet of 30-pound minimum basis weight kraft paper conforming to type I grade B of UU-P-268. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that inspection shall be in accordance with 4.4.3.

5.2.2.1 Weather-resistant fiberboard containers. When specified (see 6.2), the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.3.

5.2.3 Commercial packing. Straps, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Palletization. When specified (see 6.2), straps, packed as specified in 5.2, shall be palletized in accordance with load type I of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

5.4 Marking. In addition to any special marking required by the contract or purchase order, shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129. Each tenth interior pack shall have attached a manila colored paper shipping tag for identification information.

6. NOTES

6.1 Intended use. The strap is intended for use in securing equipment to the M-1972 pack frame assembly.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number and date of this document.
(b) When a first article sample is required (see 3.2, 4.3 and 6.4).
(c) Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
(d) When weather resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
(e) When palletization is required (see 5.3).

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6.3 Samples. For access to samples (see 3.1), address the procuring activity issuing the invitation for bid.

6.4 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of DAR 7-104.55. The first article should be a preproduction sample consisting of one completed strap. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for inspection and approval of the first article.

6.5 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the document (see 3.3).

6.6 As a matter of information only, sources of supply for the fastener and hook specified in 3.3.3 and 3.3.4 are:

(a) Arizona Gear and Manufacturing Co.
3544 East Fort Lowell Road
P.O. Box 6089
Tucson, AZ 85716

(b) Novelty Products Company
40 Washington Avenue
Bellevielle, NJ 07109

(c) G.E. Prentice Manufacturing Co.
Kensington, CT 06037

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - NU

Review activities:

Army - MD
Navy - MC
DLA - CT

User activity:

Navy - YD

Preparing activity:

Army - GL

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-S-43828A Strap, Webbing, Cargo Tie Down, Lightweight Pack Frame M-1972

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

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